

Open Issues in `std::execution::task``



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Abstract

`std::execution::task` (P3552R3^[1]) had open issues identified by national ballot comments, LEWG issues, and published papers. Croydon resolved several. This paper classifies each issue by whether it can be resolved after C++26 ships or whether shipping forecloses the fix, and notes which classified issues were addressed at Croydon.

Revision History

R2: April 2026 (post-Croydon mailing)

- Updated classification to reflect Croydon motions 28-29, 33, 35-38. Acknowledged issues resolved by P3941R4, P3927R2, P3980R1, and P4151R1. Rewrote allocator descriptions to credit P3980R1 while preserving residual frame-allocator concerns.

- Reference corrections.

R1: March 2026 (prior to Croydon meeting)

- Complete rewrite as an informational classification of open issues.

R0: February 2026 (pre-Croydon mailing)

- Original analysis of structural gaps. See [P4007R0](#)^[2].

1. Disclosure

The author provides information and serves at the pleasure of the committee.

Coroutine-native I/O and `std::execution` are complementary. Each serves the domain where its design choices pay off.

The authors developed [P4007R0](#)^[2] ("Senders and Coroutines") and [P2583R3](#)^[3] ("Symmetric Transfer and Sender Composition"). The classification below holds regardless of whether any alternative design exists.

This paper asks for nothing.

2. Fixed After Ship

`task`'s `promise_type` is a class template instantiated in user code. Its `operator new`, allocator selection, stop token storage, environment forwarding, and destruction ordering can change between standard revisions without binary incompatibility. These issues are fixable post-ship: [Unusual Allocator Customisation](#)^[4], [Flexible Allocator Position](#)^[4], [Shadowing The Environment Allocator](#)^[4], [Stop Source Always Created](#)^[4], [Stop Token Default Constructible](#)^[4], [Task Not Actually Lazily Started](#)^[4], [Frame Destroyed Too Late](#)^[4], [No Default Arguments](#)^[4], [unhandled_stopped Not noexcept](#)^[4], [Environment Design Inefficient](#)^[4], [Non-Sender Awaitables Unsupported](#)^[4], [Future Language Feature Could Avoid `co\_yield`](#)^[4], [No TLS Capture/Restore Hook](#)^[4], [return_value/return_void Have No Specification](#)^[4], [co_return { args... } Unsupported](#)^[4], [change_coroutine_scheduler Requires Assignable Scheduler](#)^[4], [Sender-Unaware Coroutines Cannot `co\_await` a task](#)^[4], [Missing Rvalue Qualification](#)^[4], [Parameter Lifetime Is Surprising](#)^[5], [No Protection Against Dangling References](#)^[5], [co_yield with_error Is Clunky](#)^[5], [co_await schedule\(sch\) Is an Expensive No-Op](#)^[5], [Coroutine Cancellation Is Ad-Hoc](#)^[5].

Of these, Croydon resolved six: Unusual Allocator Customisation, Flexible Allocator Position, and Shadowing The Environment Allocator were addressed by [P3980R1](#)^[6]; `unhandled_stopped Not noexcept`, `change_coroutine_scheduler` Requires Assignable Scheduler (the mechanism was removed entirely), and Missing Rvalue Qualification were addressed by [P3941R4](#)^[13]. The remaining issues are still open and fixable post-ship.

`affine` semantics (formerly `affine_on`, renamed by [P4151R1](#)^[14]), rescheduling behavior, and algorithm customization are specification-level concerns. Tightening requirements or adding default implementations does not change any published interface. These issues are fixable post-ship: [affine_on Default Implementation Lacks Specification](#)^[4], [affine_on Semantics](#)

Not Clear^[4], `affine_on` Shape May Not Be Correct^[4], `affine_on` Shouldn't Forward Stop Requests^[4], `affine_on` Customisation For Other Senders^[4], Starting a `task` Reschedules Unconditionally^[4], Resuming After a `task` Reschedules Unnecessarily^[4], `bulk` vs. `task_scheduler`^[4], No Completion Scheduler^[4], `with_awaitable_senders` Unused^[4].

Of these, Croydon resolved eight. P3941R4^[13] rewrote scheduler affinity from scratch - `affine_on` was made unary, `change_coroutine_scheduler` removed, and `get_start_scheduler` introduced - resolving the five `affine_on/affine` items and both rescheduling items. P4151R1^[14] renamed `affine_on` to `affine`. P3927R2^[15] resolved `bulk` vs. `task_scheduler` by giving `task_scheduler` parallel bulk support. Two items remain open: No Completion Scheduler and `with_awaitable_senders` Unused.

3. Not Fixable Post-Ship

The issues in this section are items where shipping forecloses the fix.

Issue	References	Fixed
Allocator Timing	P3980R1 ^[6] , P3796R1 ^[4] , LWG 4356 ^[7] , US 254-385 ^[8]	partial
Allocator Propagation	P3980R1 ^[6] , P3796R1 ^[4]	partial
Error Return	P3950R0 ^[9] , P3801R0 ^[5] , P1713R0 ^[10]	no
Symmetric Transfer	P2583R3 ^[3] , US 246-373 ^[11] , LWG 4348 ^[12] , P3801R0 ^[5] , P3796R1 ^[4]	no

- **Allocator Timing.** P3980R1^[6] separates frame allocation from environment allocation: the environment allocator is now sourced from `get_allocator(get_env(rcvr))` at `connect` time, resolving environment-based injection. The frame allocator, however, remains call-site-specified via `allocator_arg` in the coroutine parameter list. This is a structural consequence of coroutine allocation timing - the frame is allocated by `operator new` before `connect/start` runs, so the receiver's environment is unavailable. Shipping standardizes the two-tier split. A design where frame allocation participates in environment-based propagation is foreclosed without a language change to coroutine allocation.
- **Allocator Propagation.** P3980R1^[6] provides environment allocator propagation: each `task` obtains its environment allocator from `get_allocator(get_env(rcvr))`, so a parent's environment allocator flows to children through the receiver chain. Frame allocator propagation remains absent. Each coroutine call site must independently specify `allocator_arg` to use a non-default frame allocator. In a deep coroutine call tree where frame allocation matters - arena-per-request, pool allocators, device memory - every function signature must carry `allocator_arg` explicitly. Shipping this design forecloses automatic frame allocator propagation through the coroutine call tree.
- **Error Return.** `task` requires `co_yield with_error(e)` to propagate an error to the caller. `co_return` cannot carry an error value because `return_value` and `return_void` are mutually exclusive in the current coroutine specification. Shipping this interface locks in the `co_yield` mechanism and forecloses `co_return`-based error propagation, which would require a language change.
- **Symmetric Transfer.** The completion functions (`set_value`, `set_error`, `set_stopped`) and `start()` return `void`, providing no channel to propagate a `coroutine_handle<>`. When a sender completes synchronously, the receiver calls `handle.resume()`

on the caller's stack, adding a frame per completion with no upper bound. Shipping this protocol forecloses the `coroutine_handle<>`-returning completion protocol that would enable symmetric transfer.

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References

Papers, issues, and ballot comments referenced in this document.

WG21 Papers

- [1] [P3552R3](#) - "Add a Coroutine Task Type" (Dietmar Kühl, Maikel Nadolski, 2025).
- [2] [P4007R0](#) - "Senders and Coroutines" (Vinnie Falco, Mungo Gill, 2026).
- [3] [P2583R3](#) - "Symmetric Transfer and Sender Composition" (Mungo Gill, Vinnie Falco, 2026).
- [4] [P3796R1](#) - "Coroutine Task Issues" (Dietmar Kühl, 2025).
- [5] [P3801R0](#) - "Concerns about the design of `std::execution::task`" (Jonathan Müller, 2025).
- [6] [P3980R1](#) - "Task's Allocator Use" (Dietmar Kühl, 2026).
- [7] [LWG 4356](#) - "`connect()` should use `get_allocator(get_env(rcvr))`".
- [8] [US 254-385](#) - C++26 NB ballot comment.
- [9] [P3950R0](#) - "`return_value` & `return_void` Are Not Mutually Exclusive" (Robert Leahy, 2025).
- [10] [P1713R0](#) - "Allowing both `co_return`; and `co_return` value; in the same coroutine" (Lewis Baker, 2019).
- [11] [US 246-373](#) - C++26 NB ballot comment.
- [12] [LWG 4348](#) - "task doesn't support symmetric transfer".

Croydon Papers (R2)

- [13] [P3941R4](#) - "Scheduler Affinity" (Dietmar Kühl, 2026).
- [14] [P4151R1](#) - "Rename `affine_on`" (Robert Leahy, 2026).
- [15] [P3927R2](#) - "task_scheduler Bulk Execution" (Eric Niebler, 2026).